## 1 Acquire

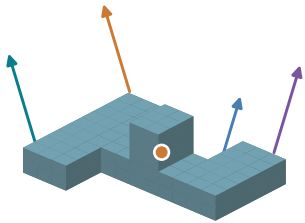


RGB-D surface and pose

$\mathcal{S},\ \hat{p}(t),\ \hat{R}(t),\ b_i$

12-view scan; calibrated mounts

## 2 Identify

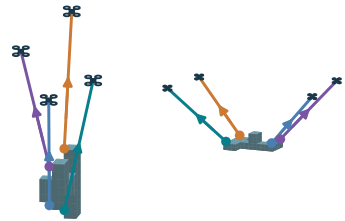


Tensions and integral momentum

$A\theta = b, \quad \theta = B\mu, \quad \mu \geq 0$

$\hat{\theta}:\ \hat{m},\ \hat{c},\ \hat{J}_O$

## 3 Reconfigure



Cable forces set support geometry

$f_i \mapsto x_{i,r}(t)$

Side-on and spread formations

**GIVEN**

Known obstacles and smooth payload pose references

**GEOMETRY**

Surface and attachments

**Constrained cable-force optimization**

$\min \mathcal{J}(f) \quad \text{s.t.} \quad G(R_r)f = Y_d\hat{\theta}$

Aperture, loading, actuation and endpoint separation

**EXECUTION**

Endpoint / attitude control

50 Hz motor commands

10 Hz RGB-D requests and synchronized tension feedback